Momentum and Impulse

Answer key

Conventions used throughout

• **Momentum (from the lesson):** $\vec{p} = m\vec{v}$, in kg·m/s. It is a vector, in the direction of the velocity.

• **Impulse (from the lesson):** $\vec{J} = \vec{F}\Delta t$, and impulse equals the *change* in momentum: $\vec{F}\Delta t = \Delta\vec{p} = m\vec{v}_2 - m\vec{v}_1$.

• **Signs:** each question names its + direction. Every velocity carries its sign in the working ($v_1 = -43$ m/s). The direction is written in square brackets only in the final answer.

• **Solve in letters first:** each unknown is isolated as a formula before any numbers go in.

• **Final answers:** on a chosen axis, the scalar with its sign and then the vector, for example $F = -4.2 \times 10^3$ N and $\vec{F} = 4.2 \times 10^3$ N [backward].

• Speeds in km/h are changed to m/s (÷ 3.6). Every step keeps full precision; only final answers are rounded.

IMPULSE

The connection between impulse and momentum

You will now work out the connection between impulse and momentum.

a) You know that according to Newton's 2nd law, $\vec{F}_{net} = m\vec{a}$. Assuming that there is only one force acting on the object, sub in for Force in equation (1)

b) You now have the product of acceleration and time: $\vec{a}\Delta t$. According to the basic definition of acceleration, what is this quantity equal to? _____ Sub it in to equation (2)

c) But of course $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$. Sub this in to equation (3) and expand. What is the connection between impulse and momentum?

d) Confirm your answer for (c) by making sure the units are the same. Check with your teacher.

Equation (1) is the definition of impulse, $\vec{J} = \vec{F}\Delta t$. Each part changes its right side one step at a time.

a) sub in for Force in equation (1)

Equation (1): $\vec{J} = \vec{F}\,\Delta t$

Put in $\vec{F} = m\vec{a}$: $\vec{J} = m\vec{a}\,\Delta t$ (2)

b) According to the basic definition of acceleration, what is this quantity equal to? Sub it in to equation (2)

Acceleration is the change in velocity divided by the time, so acceleration times time is the change in velocity.

Definition: $\vec{a} = \dfrac{\Delta \vec{v}}{\Delta t} \quad \Rightarrow \quad \vec{a}\,\Delta t = \Delta \vec{v}$

Put it into (2): $\vec{J} = m\,\Delta \vec{v}$ (3)

c) Sub this in to equation (3) and expand. What is the connection between impulse and momentum?

Put in $\Delta \vec{v} = \vec{v}_2 - \vec{v}_1$: $\vec{J} = m(\vec{v}_2 - \vec{v}_1)$

Expand: $\vec{J} = m\vec{v}_2 - m\vec{v}_1$

Use $\vec{p} = m\vec{v}$: $\vec{J} = \vec{p}_2 - \vec{p}_1 = \Delta \vec{p}$

Impulse equals the change in momentum.

d) Confirm your answer for (c) by making sure the units are the same.

Impulse is force × time. Momentum is mass × velocity. Write the newton in base units.

$$\text{Impulse:}\quad [\text{N}][\text{s}] = \left[\text{kg}\,\frac{\text{m}}{\text{s}^2}\right][\text{s}] = \text{kg}\,\frac{\text{m}}{\text{s}}$$

$$\text{Momentum:}\quad [\text{kg}]\left[\frac{\text{m}}{\text{s}}\right] = \text{kg}\,\frac{\text{m}}{\text{s}}$$

The units match: 1 N·s = 1 kg·m/s.

FINAL ANSWER

a) $\vec{J} = m\vec{a}\Delta t$

b) $\vec{a}\Delta t = \Delta\vec{v}$, so $\vec{J} = m\Delta\vec{v}$

c) $\vec{J} = m\vec{v}_2 - m\vec{v}_1 = \Delta\vec{p}$: impulse equals the change in momentum

d) N·s = kg·m/s, the same units

EXAMPLE

Change in momentum of a baseball

Infamous Toronto Blue Jay slugger Mr. Kassil is at bat. A 145 g baseball is pitched to him at 43 m/s [S]. Kassil hits the ball so that its final velocity is 65 m/s [N].

a) What is the change in momentum of the ball?

b) If Kassil's bat was in contact with the ball for 0.035 s, how much force did he exert on the ball?

GIVEN

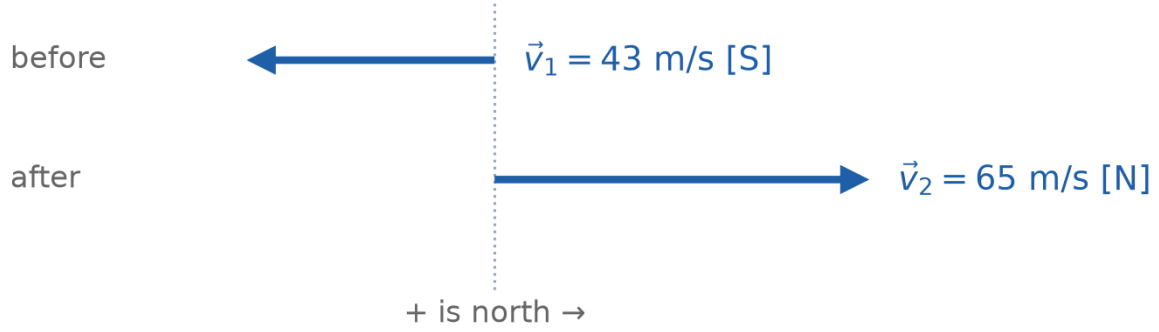
$m = 145\ \text{g} = 0.145\ \text{kg}$ $\qquad$ $v_1 = -43\ \text{m/s}$

$v_2 = +65\ \text{m/s}$ $\qquad$ $\Delta t = 0.035\ \text{s}$

$\Delta p = ?$ $\qquad$ $F = ?$

Sign convention: let north be positive (as in the worksheet's solution), so 43 m/s [S] is −43 m/s.

The ball's velocity before and after the hit



a) What is the change in momentum of the ball?

Change means final minus initial. The initial velocity is negative, so subtracting it adds.

Formula: $\Delta p = p_2 - p_1 = mv_2 - mv_1$

Put in the numbers: $\Delta p = (0.145)(+65) - (0.145)(-43)$

$\Delta p = 9.425 - (-6.235) = +15.66$ kg·m/s

Δp is positive, so the change in momentum points north.

b) If Kassil's bat was in contact with the ball for 0.035 s, how much force did he exert on the ball?

Impulse equals the change in momentum. Isolate F in letters first.

Formula: $F\,\Delta t = \Delta p$

Divide both sides by Δt: $F = \dfrac{\Delta p}{\Delta t}$

Put in the numbers: $F = \dfrac{15.66}{0.035} = +447.4\ \text{N}$

FINAL ANSWER

a) Δp = +16 kg·m/s

$\vec{\Delta p}$ = 16 kg·m/s [N]

b) F = +4.5 × 10^2 N

$\vec{F}$ = 4.5 × 10^2 N [N]

Worksheet answer: a) 16 kg·m/s [N]; b) 447 N [N] (the worksheet's solution on page 4)
Why the answers differ: b) Same value, different rounding. 15.66 ÷ 0.035 = 447.4 N. The data (43 m/s, 65 m/s, 0.035 s) have 2 significant figures, so the key gives 4.5 × 10^2 N; the worksheet keeps 3.

APPLYING IMPULSE

Airbags

Imagine a car collision in which the driver's body flies forward and hits the steering wheel. Let's say the upper body has a mass of 30.0 kg and the car was travelling at 60.0 km/h. The time taken for the body to be brought to rest by the steering wheel is 0.01 s because the steering wheel is so hard.

a) Use concepts of impulse and momentum to determine the force exerted on the person's body by the steering wheel.

b) Now say the car has airbags so the time taken to stop is 0.12 s. The initial and final speeds are the same. Determine the force in this case.

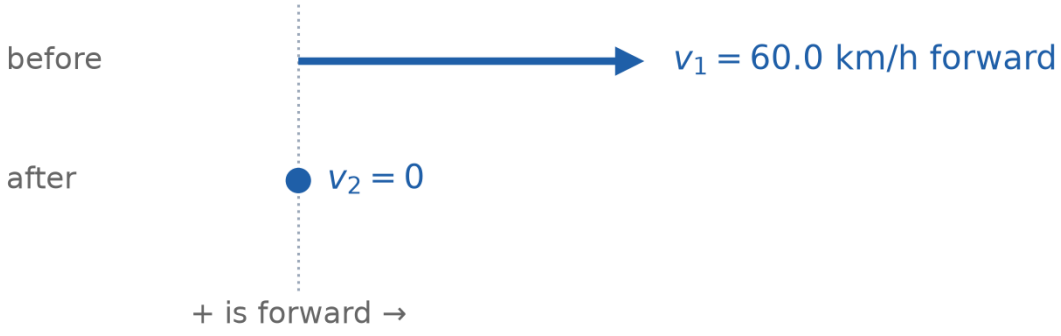
c) Explain, in words, why airbags and so important using the concepts of impulse and momentum. Can you think of other situations where this principle would apply?

GIVEN

$m = 30.0$ kg	$v_1 = 60.0$ km/h	$v_2 = 0$
$\Delta t_a = 0.01$ s	$\Delta t_b = 0.12$ s	$F = ?$

Sign convention: + is forward, the direction the body is moving.

The body's velocity before and after



a) Use concepts of impulse and momentum to determine the force exerted on the person's body by the steering wheel.

Step 1 – Change the speed to m/s

Divide by 3.6: $v_1 = \dfrac{60.0}{3.6} = 16.667 \text{ m/s}$

Step 2 – Change in momentum

The body is brought to rest, so $v_2 = 0$.

Formula: $\Delta p = mv_2 - mv_1$

Put in the numbers:

$$\Delta p = (30.0)(0) - (30.0)(16.667) = -500.0 \text{ kg·m/s}$$

Step 3 – Force from the steering wheel

Formula: $F\,\Delta t = \Delta p \quad \Rightarrow \quad F = \dfrac{\Delta p}{\Delta t}$

Put in the numbers: $F = \dfrac{-500.0}{0.01} = -50000 \text{ N}$

F is negative: the force on the body points backward, against its motion.

b) Now say the car has airbags so the time taken to stop is 0.12 s. The initial and final speeds are the same. Determine the force in this case.

The speeds are the same, so the change in momentum is the same −500.0 kg·m/s. Only the time changes.

Formula: $F = \dfrac{\Delta p}{\Delta t}$

Put in the numbers: $F = \dfrac{-500.0}{0.12} = -4166.7 \text{ N}$

c) Explain, in words, why airbags and so important using the concepts of impulse and momentum. Can you think of other situations where this principle would apply?

The body's change in momentum is the same with or without the airbag, so the impulse $F\Delta t$ needed to stop it is the same. The airbag makes the stopping time 12 times longer, so the force is 12 times smaller, small enough to avoid serious injury. The same idea, a longer time for a smaller force, is used in bending your knees when you land from a jump, padded helmets and gym mats, crumple zones in cars, and pulling your hand back when you catch a hard ball.

FINAL ANSWER

a) $F = -5 \times 10^4$ N

$\vec{F} = 5 \times 10^4$ N [backward]

b) $F = -4.2 \times 10^3$ N

$\vec{F} = 4.2 \times 10^3$ N [backward]

c) Same Δp, longer stopping time, so a smaller force

Worksheet answer: a) -5.0×10^4 N; b) -4.2×10^3 N
Why the answers differ: a) Same value, different rounding. $-500.0 \div 0.01 = -50\,000$ N. The time (0.01 s) has only 1 significant figure, so the key gives -5×10^4 N; the worksheet writes 2 figures.

CONCEPT QUESTION 1

Small force, large impulse

1. It is possible for an object to receive a larger impulse from a small force than from a large force? Explain.

Impulse depends on the time as well as the force.

Formula: $\vec{J} = \vec{F}\,\Delta t$

Example:

$(10\text{ N})(5.0\text{ s}) = 50\text{ N·s}$ but $(100\text{ N})(0.10\text{ s}) = 10\text{ N·s}$

The small force gives the larger impulse because it acts for a much longer time.

FINAL ANSWER

Yes: a small force that acts for a long enough time gives a larger impulse (J = FΔt)

CONCEPT QUESTION 2

Putty and rubber ball

2. A piece of putty dropped vertically to the floor sticks and stops whereas a rubber ball dropped from the same height bounces until it reaches almost the same vertical height. Which has the greater change in momentum during the collision with the floor?

Take + up. Both hit the floor with the same velocity, −v (same drop height). To compare them fairly, take them to have the same mass m.

The putty stops, so its final velocity is 0. The ball leaves the floor at almost +v, because it rises to almost the same height.

Formula: $\Delta p = mv_2 - mv_1$

Putty: $\Delta p = m(0) - m(-v) = +mv$

Rubber ball: $\Delta p \approx m(+v) - m(-v) = +2mv$

The ball's momentum has to be stopped and then reversed, so its change is almost twice as big.

FINAL ANSWER

The rubber ball: about 2mv, compared with mv for the putty (for equal masses)

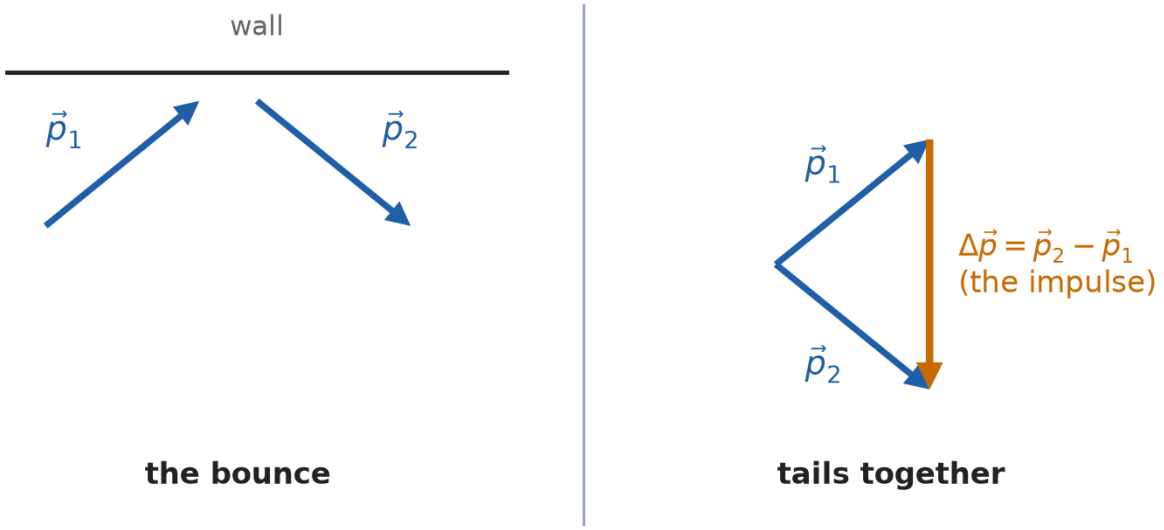
CONCEPT QUESTION 3

Impulse on a bouncing ball

3. A ball bounces off a wall as shown below. The vectors represent the momentum of the ball. Draw a vector that represents the direction of the impulse of the ball.

Impulse equals the change in momentum, $\Delta\vec{p} = \vec{p}_2 - \vec{p}_1$. To subtract vectors, draw both from the same point. $\Delta\vec{p}$ is the arrow from the tip of $\vec{p}_1$ to the tip of $\vec{p}_2$.

The bounce, and the two momentum vectors drawn tail to tail



the bounce **tails together**

The parts of $\vec{p}_1$ and $\vec{p}_2$ along the wall are equal, so they cancel. Only the part toward the wall reverses.

FINAL ANSWER

The impulse points straight away from the wall, perpendicular to it (straight down in the worksheet's picture). That is the direction of the wall's force on the ball.

CONCEPT QUESTION 4

Two playdough balls

4. Two playdough balls rolling towards each other collide, stick together and stop. Is the momentum of the system conserved? (Momentum is conserved if the initial total momentum is equal to the final total momentum.) Explain.

Yes. After the collision nothing is moving, so the final total momentum is zero. The balls rolled toward each other, so their momenta were in opposite directions, and for them to stop dead those momenta must have been equal in size: the initial total was also zero. During the collision each ball pushes on the other with an equal and opposite force for the same time, so the impulses cancel and the total does not change.

Before: $\vec{p}_{tot} = \vec{p}_1 + \vec{p}_2 = 0$ (equal sizes, opposite directions)

After: $\vec{p}_{tot}{}' = (m_1 + m_2)(0) = 0$

FINAL ANSWER

Yes: the total momentum is zero before and zero after. (Kinetic energy is not conserved; it becomes thermal energy and deformation.)